

# Supplementary Material A: Replication Analysis of Gruau (1994) — Architecture Learnability and the Weight-Reset Protocol

## 1 Overview

This supplement reports a replication and extension of Gruau’s cellular encoding framework (Gruau, 1994), which demonstrated that tree-structured genomes can develop into functional neural networks through a growth process. Our replication confirms Gruau’s core mechanism and suggests a limitation in the original fitness evaluation protocol, motivating the weight-reset evaluation used in the main paper.

## 2 Gruau’s Cellular Encoding

Gruau (1994) introduced *cellular encoding*, where a tree-structured genome encodes a developmental program. Each node in the tree is an instruction (e.g., SEQuential division, PARallel division, weight modification) that operates on a growing graph of neurons. The genotype is a compact program; the phenotype is the resulting neural network.

The key insight of cellular encoding is the separation of genotype (developmental program) from phenotype (instantiated network). A single genome can produce networks of varying sizes and connectivity patterns depending on the developmental context. Crucially, this means the same genome can produce *different weight initializations* across multiple developmental runs—a property we exploit in our architecture-first evaluation protocol.

## 3 Replication Method

We replicate Gruau’s cellular encoding for the XOR task, using:

- Tree-structured genomes with SEQ, PAR, and weight-assignment nodes.
- A graph-rewriting developmental process that maps genome  $\rightarrow$  network.
- Tournament selection with subtree crossover and node mutation.
- Population size 50, 100 generations.

We compare two fitness evaluation protocols:

**Original (weight-based)** The developed network’s weights are used directly for forward-pass evaluation. Fitness =  $1 - \text{MSE}(\text{XOR})$ . This selects for lucky weight configurations.

**Modified (architecture-based)** After development, all weights are randomly re-initialized. The network is trained via SGD for  $N$  steps. Fitness = post-training accuracy. This selects for architectural learnability.

Table 1: Gruau replication results: architecture learnability under two fitness protocols.

| Protocol                | Architecture Convergence | Mean Final MSE | Best MSE |
|-------------------------|--------------------------|----------------|----------|
| Original (weight-based) | 0/20 (0%)                | —              | 0.064    |
| Modified (learnability) | 10/10 (100%) evolved     | 0.075          | 0.064    |
|                         | 0/20 (0%) random         | 0.195          | 0.195    |

## 4 Results

**Finding 1: Original fitness does not select for architecture.** When fitness measures only forward-pass accuracy on fixed weights, 0% of evolved architectures encode learnability. The architecture is a *genetic hitchhiker*—it happens to be present in individuals with good weights, but the selection pressure never tests the architecture’s contribution to learning.

**Finding 2: Modified fitness selects for learnable architectures.** When fitness measures post-training accuracy after weight re-initialization, 100% of evolved architectures (10/10 trials) successfully encode learnability, compared to 0% for randomly generated architectures (0/20) under the same protocol. The evolved architectures achieve  $2.6\times$  lower MSE than random architectures.

## 5 The Best Evolved Architecture

The best architecture evolved under the learnability protocol (MSE = 0.064, accuracy = 100%) has the following structure:

- 5 neurons total (2 input, 2 hidden, 1 output)
- 5 synapses
- Hidden units: neuron 3 (GELU, bias  $-0.916$ ), neuron 4 (GELU, bias  $+0.962$ )
- Key synapses: input $\rightarrow$ hidden weights  $[-2.461, +2.646]$ , hidden $\rightarrow$ output weights  $[+8.049, +0.629]$ , and an inter-hidden connection  $[+0.899]$

The presence of an inter-hidden connection (neuron 3  $\rightarrow$  neuron 4) indicates that evolution discovered a recurrent solution to XOR under the learnability constraint, despite XOR being a feedforward-solvable problem. The asymmetry in output weights (8.049 vs 0.629) suggests a main signal path with a corrective branch—a structure that would not emerge under pure weight optimization.

## 6 Implications for the Main Paper

This replication establishes three foundations for the architecture-first philosophy used in the main paper:

1. **Architectural knowledge is selectable.** When fitness demands learnability (not just current performance), evolution can discover architectures whose loss landscapes are shaped for gradient-based learning. This validates the weight-reset evaluation protocol used throughout our experiments.
2. **The distinction between parametric and architectural knowledge is empirically grounded.** The same genome evaluated under weight-based fitness produces no learnable architectures; evaluated under architecture-based fitness, it reliably produces them. The difference is not in the genome representation—it is in what the selection pressure demands.

3. **Minimal genomes suffice.** A 5-neuron, 5-synapse network solves XOR with perfect accuracy after SGD training. This supports our approach of evolving a minimal set of architectural parameters  $(\rho, \gamma, s)$  rather than full connectivity patterns.

## 7 Reproducibility

The complete replication data is available in the repository as `gruau_result.json`. The file contains:

- The best-evolved genome tree structure
- The developed network architecture (neurons, synapses, activations)
- Full training history (MSE per generation)
- Random baseline comparison data

The replication code is integrated into the project repository and can be executed via the Docker setup described in the main paper’s README.

## References

Gruau, F. (1994). *Neural Network Synthesis Using Cellular Encoding and the Genetic Algorithm*. PhD Thesis, École Normale Supérieure de Lyon.